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Research Presentation and Proposal

Compound Flood Modelling, Bayesian Data Assimilation, Production-scale Continental Modeling

Selected work across four research and applied one case study, spanning the full hazard – exposure – risk pipeline.

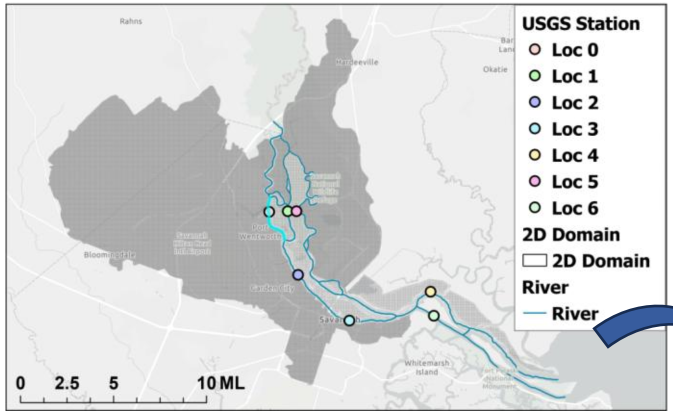
Hydrology | Flood risk modelling | Bayesian + ML methods | Fluvial, urban flooding

August 2026

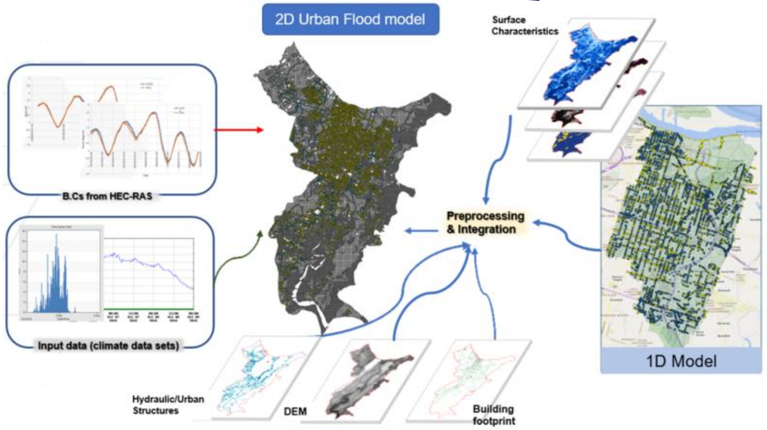
Physical Modeling, ML Surrogates, and Remote Sensing: A Compound Flood Case Study

Project Overview

- Goal: Quantify how flood drivers interact. to produce urban flooding — and determine which driver controls where
- Objectives:
  1. Build and validate a coupled hydrodynamic model capturing surface and drainage-network flow for 1233 km^2 urban watershed
  2. Quantify driver interactions across a designed scenario ensemble
  3. Verify predicted extent independently against satellite observation
  4. Suggest a methodology to quantify uncertainty and decrease it
- Methods:
  1. Construction of coupled HEC-RAS-PCSWMM flood model for compound urban flood with flexible resolution to save computational cost.
  2. Validated against Hurricanes Matthew, Irma, Dorian, Idalia at five USGS gauges
  3. Latin Hypercube ensemble scenario; PLSR / Random Forest / XGBoost surrogates
  4. Independent verification: Sentinel-1 SAR
  5. Application of Bayesian Data Assimilation (to combine physical model as prior and measurement as likelihood)



One-way coupling



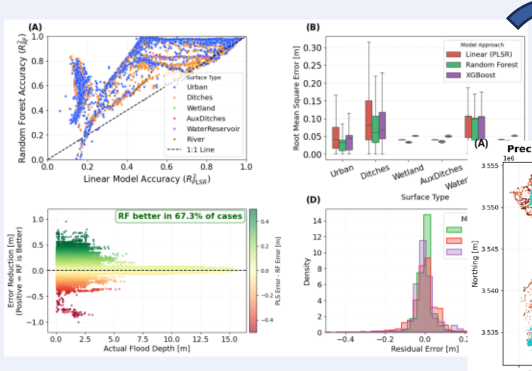
WHAT THIS DEMONSTRATES

Compound flood modeling, coupled modeling, Bayesian Data Assimilation, ML based flood driver analysis

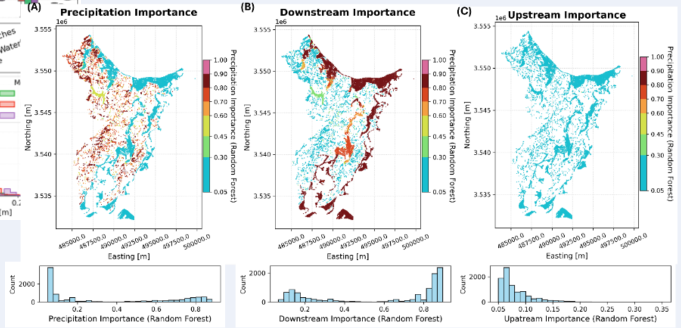
Physical Modeling, ML Surrogates, and Remote Sensing: A Compound Flood Case Study

- Key Results:
  1. Spatially heterogeneous response of water depth, residual time to flood drivers (precipitation, upstream condition, surge)
  2. Regarding fluvial flood, spatial correlation is maintained through 4 different hurricanes (Matthew, Irma, Dorian, Idalia)
  3. FEMA Zone is weighted to one driver (surge), while results show Zone X is more influenced by precipitation than surge.
  4. Processed SAR showed a good validation material, suitable for rural areas, not for urban areas with limitations
  5. Possibility of an optimization of flood network. Only 3 of 5 stations are necessary, proved by Bayesian data assimilation (DA)

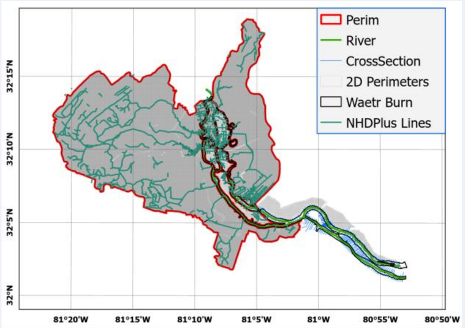
Flood Driver Analysis



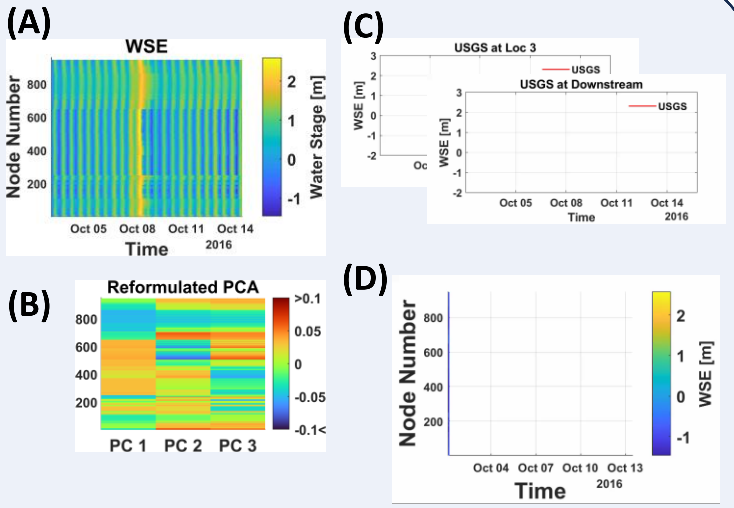
Surrogate model using PLSR, FR, XGBoost



Data Assimilation

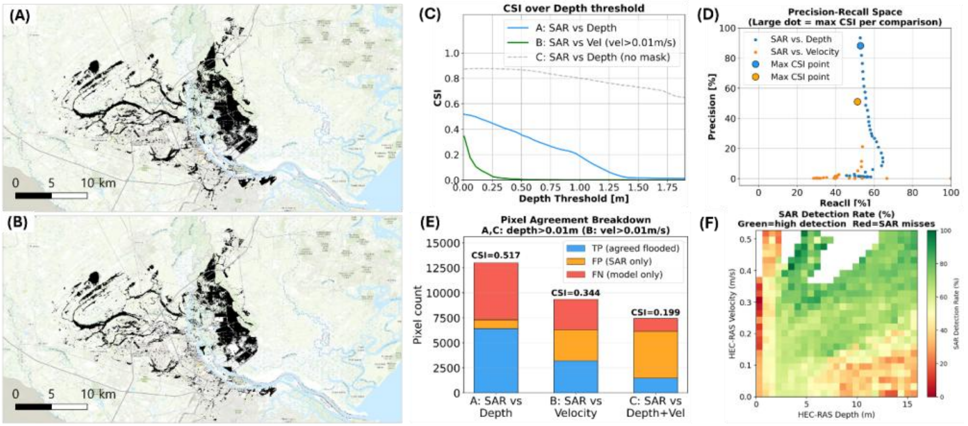


Combining physical model as prior and measurement as likelihood

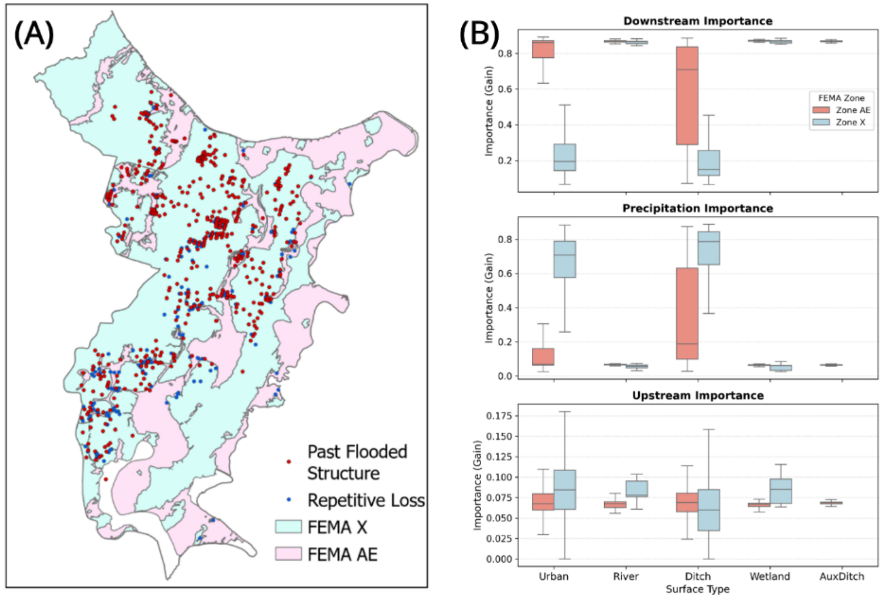


Section 1: Relation of Project to CIROH's modeling research

Component	Connection to CIROH Hydrology Modeling
Coupled 2-D hydraulic modeling	Optimal method for operational compound flood modeling
Surrogates for driver attribution	Identification of forecast uncertainty in riverine-coastal system; spatial distribution of driver dominance
Verification with remote sensing data and global data product	Remote-sensing evaluation of model skill, and identification of where the benchmark fails
Bayesian data assimilation	Fusing sparse observations with physical model priors to reconstruct flood state in data-sparse domains
Limited application of FEMA zone in compound flood area	risks in using existing flood-risk datasets for operational assessment



Characteristics of SAR compared to flood model results (possible failure at extreme condition (too shallow and too fast), Froude-based assumption



Different dominance of flood drivers to FEMA zone

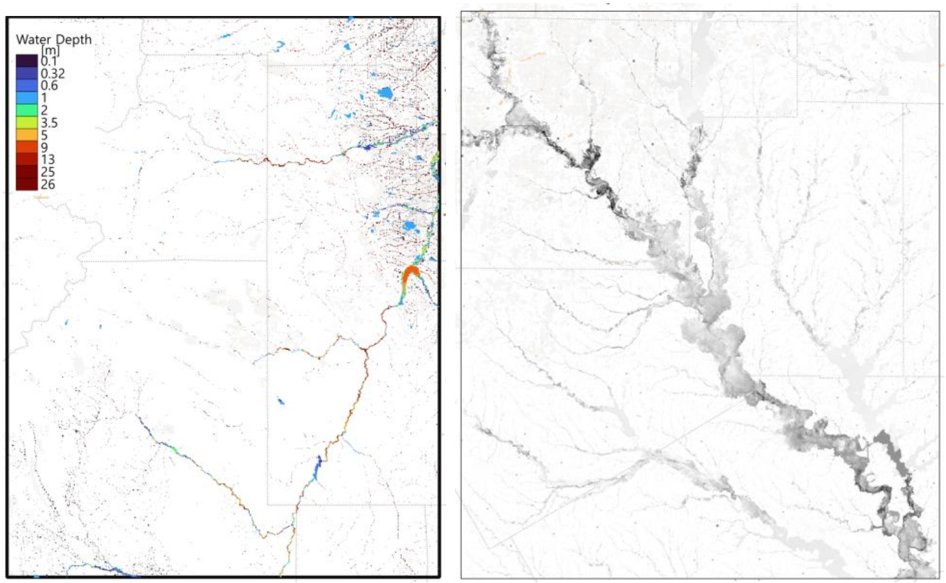
AI/ML — PLSR, Random Forest, XGBoost surrogates on physical model output; tree-based regionalization of return-period discharge at unmeasured boundary points

Large-domain hydrologic modeling — pluvial/fluvial flood modeling, patches of ~75 × 110 km (~66,000 km^2), global operational flood program; MERIT DEM and MERIT Hydro; GloFAS / EFAS as reference

High-performance computing — HEC-RAS on Linux HPC, one core per simulation, AWS-backed data management

Programming and open software development — public repositories for the coupled model, driver analysis, and PIGA; automated HEC-RAS execution with HDF file control

Research studies applying hydrologic models — validated compound flood modeling; PIGA (*J. Hydrology*, 2024); Pipelined model construction for large hydrology model.



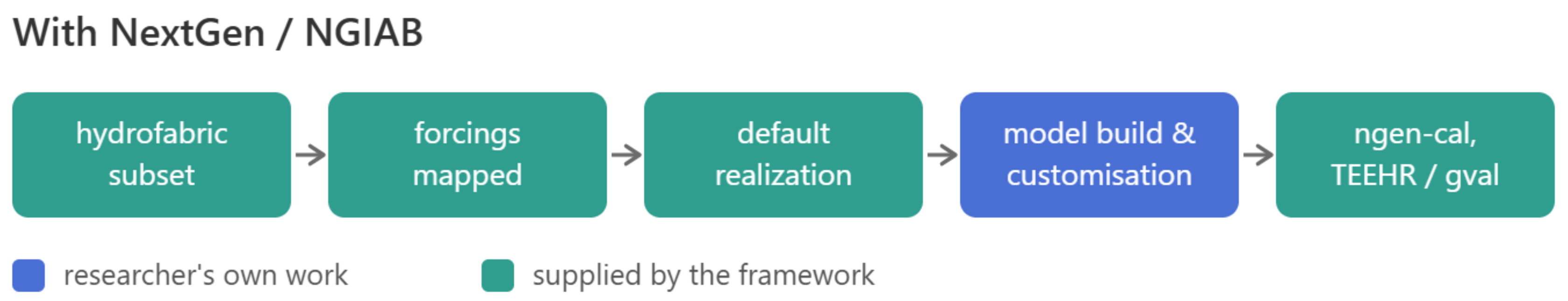
* Large Domain Model example: 1 in 100 yr pluvial flood simulation + 1 in 100 yr fluvial flood simulation for Denver CO (left), Dallas TX (right) of 75 km by 110 km grid

Model setting: grid-based tiling, MERIT DEM, MERIT Hydro
Parameter estimation: tree-based ML (Random Forest, XGBoost)
Solver: HEC-RAS (mesh by Windows GUI) under Linux HPC
AWS for data management, GitHub code repository
Validation: GloFAS and EFAS, Deltares, and FEMA

Section 2: Understanding of NextGen Water Resource Modeling

A framework, not a model — it composes models rather than computing water balance itself.

- **BMI 2.0:** model-agnostic interface (initialize, update, get_value, set_value, finalize). The bridge that makes models swappable and couplable.
- **Hydrofabric** (OGC HY_Features): divides, flowpaths, nexus points. The shared spatial addressing scheme.
- **Model formulations:** CFE (conceptual rainfall-runoff), Noah-OWP-Modular (land surface fluxes), TOPMODEL, CASAM, LSTM.
- **Process modules:** PET, SoilMoistureProfiles, SoilFreezeThaw.
- **t-route:** 1-D network routing; includes reservoir modules and streamflow DA against USGS gauges.
- **NGIAB:** containerized deployment; plugin-based model integration without editing the build.
- **NRDS:** community-editable CONUS data stream.
- **Evaluation & calibration:** TEEHR, gval; ngen-cal.



Section 2: Current and Contributable research topics

Research that has used NextGen

Research Topic	Applied / Recent Study
Data Assimilation	Ensemble Kalman filter module assimilation of in-situ streamflow into CFE
Coastal coupling	Inclusion of GeoClaw initiated in 2022.
ML surrogate for FIM	HAND-FIM correction via HEC-RAS results: −18% false alarms, +26% CSI *(EM&S 194, 2025)*

Ideas to advance the framework

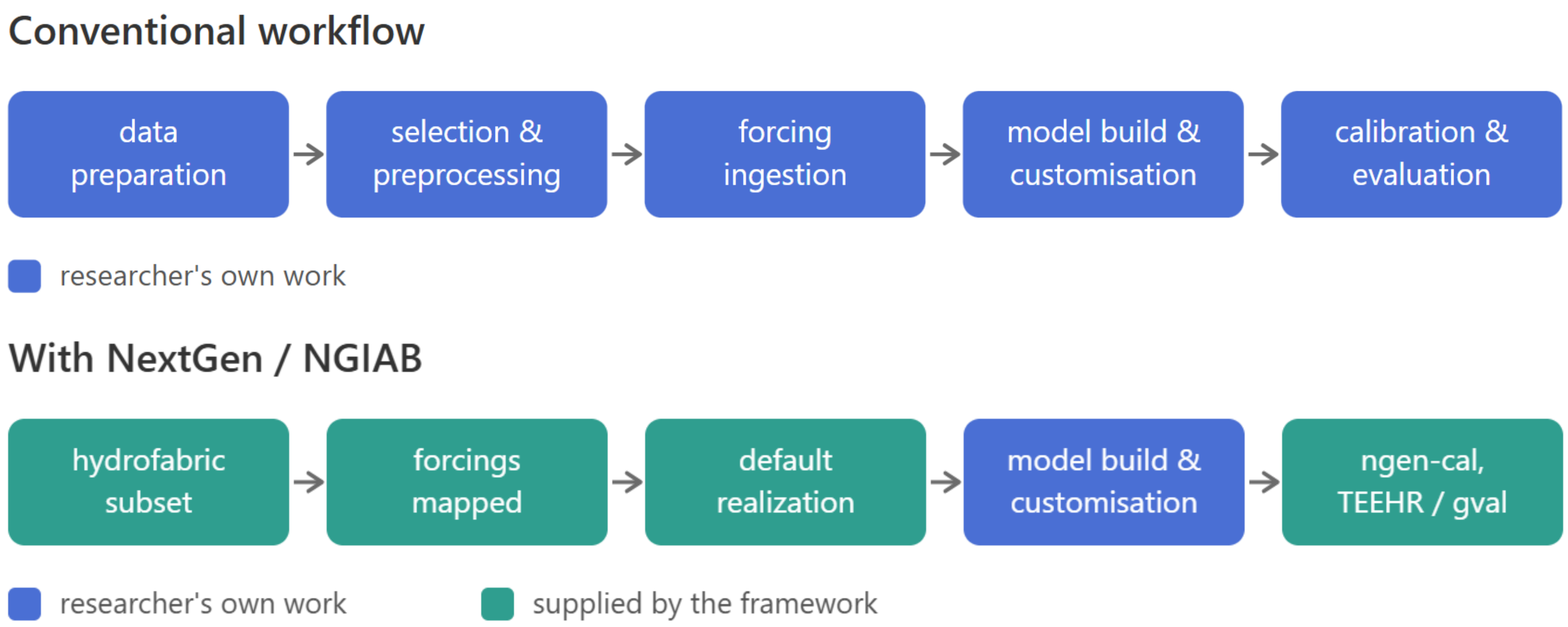
Gap	Opportunity
Data Assimilation is gauge only	Possible inclusion of geospatial approach of remote sensing data
BMI-native 2-D hydraulic model	Conventional 2D surface routing model could be added; GeoClaw for coastal area, not floodplain
Supplement to FIM	FIM is based on geography not physical model, effective post processing. Possible disadvantages under highly dynamic conditions
Addition of urban flood model	A pipe-network formulation (SWMM-class) for pluvial and drainage-controlled flooding

Section 3: Engaging researchers with the NextGen Framework

Q: Why Should Researchers Use NextGen?

A: The practical cost of hydrologic modeling isn't the science only— it's also the setup.

- Save the setup time — no separate effort for input preparation or boundary condition generation
- Start from a working configuration — forcings, hydrofabric, and defaults are already supplied
- A paved path — calibration (ngen-cal), evaluation (TEEHR, gval)
- Low barrier by design — containerized via NGIAB; adding a model needs no build-system work
- One entry point for many disciplines — hydrologists, ML researchers, and software developers build on the same foundation



A shared framework enables research questions that no single project can address.

Model combination becomes a tractable research question — different formulations may produce different results. So optimized combination of models can be achievable more easily than before.

Process-based and machine learning models compete on equal terms — an LSTM and a conceptual model are both formulations; the framework privileges neither

A continental reference that already runs — NRDS provides a continuously updated CONUS baseline against which any study may be measured.

Model comparisons become meaningful — a common hydrofabric, common forcings, and common evaluation tools enables comparison with less uncertainty due to process selection

Research reaches operations — a defined research-to-operations pathway toward the National Water Model because NEXTGen is the architecture for transition.

The direct benefit to the contributor.

- **Continental-scale validation at no cost** — a contributed configuration is regularly > reliable boundary condition
- **Rapid identification of failure modes** — many users across many basins surface edge cases that a single-basin study would not reveal
- **Contributions become infrastructure** — work that others build upon rather than read once
- **Longevity beyond the funding period** — a contributed configuration continues running after a project closes
- **A low-friction submission process** — NRDS accepts parameter updates through a documented issue template, not a proposal cycle
- **A path to operational use** — a defined route from contribution toward operational forecasting

Physics-Constrained Machine Learning for Flood Inundation in NextGen

- Motivation: FIM is an efficient post-processing method: NWM discharge is converted to inundation extent through a synthetic rating curve built on the uniform-flow assumption. That assumption holds where flow is channel-confined and fluvially driven. It degrades as floodplains engage and downstream controls activate — conditions that intensify with event magnitude. Moving toward the coast, tidal and surge control introduce backwater the method cannot represent, and compound forcing presented, since discharge is the only input.
- Solution: Add constraints from physical condition such as physics-informed machine learning.
- Goal: Determine whether, and if so, where low-complexity FIM fails, then correct it with physics-constrained machine learning trained on both simulated and observed flood extent.

Phase I — Diagnosis

1. Quantify HAND-FIM skill degradation across return period, terrain class, and downstream control
2. Establish observability-aware evaluation — condition SAR-based skill on where the benchmark is reliable

Phase II — Correction

3. Generate subgrid 2-D ensembles as training data, indexed to the hydrofabrics in NextGen framework
4. Train a physics-constrained correction converting the rating curve into a rating surface, using SAR observations as anchoring constraints

Methods

Phase I — Diagnosis

Benchmarks from existing high-fidelity models: ras2fim library, the validated Savannah model

Stratify HAND-FIM error by return period, slope class, confinement, and downstream control type

Sentinel-1 pipeline: thresholding, permanent-water masking, mesh projection, CSI/POD/FAR

Phase II — Correction

Objective: Rating curve to rating surface

- **Dense, synthetic** — subgrid 2-D ensembles across discharge × downstream stage, spanning magnitudes no observation record covers
- **Sparse, observed** — SAR-derived stage at acquisition times, weighted by the Phase I observability mask
- Ensemble discharge and downstream stage for a two-variable surface as stage = f(Q, η of downstream)
- Coastal boundary supplied by existing SCHISM total water level
- Rating curve → rating surface: stage = f(Q, η_downstream), coastal boundary from NWM's existing SCHISM total water level (or GloClow)
- Target: log-ratio of benchmark stage to rating-curve stage — multiplicative error structure
- Features from the subgrid tables: conveyance ratio (floodplain/channel), wet fraction, downstream stage index, reach slope, return period etc..
- Validation and Calibration

Anticipated Result

Phase I

1. **A skill-degradation surface**
   HAND-FIM error as a function of return period × terrain class × downstream control. Inundatuion mapping supported by hydraulics addition.
2. **A regime map with a computable criterion**
   Classification of CONUS zones into the three tiers: HAND (FIM) adequate, rating-surface correctable, 2-D required (coastal or highly dynamic areas).
3. **An observability field**
   Froude-based detectability converts a hidden assumption in SAR-based FIM evaluation into a measurable quantity — skill reported where the benchmark is reliable. Includes the legitimate finding that HAND may be sufficient across much of CONUS.

Phase II

1. **A two-variable rating surface**
   stage = f(Q, η of downstream) — physically interpretable, transferable to ungauged reaches, consuming NWM's existing SCHISM total water level.
2. **Subgrid fidelity thresholds** — the $\Delta x/\delta x$ limits at which coarse cells preserve hydraulic accuracy, stratified by terrain
3. **Extension of NextGen DA :** From gauge streamflow to spatial observation.

Summary of impact and benefit of the research

Operational

FIM coverage is nearly complete under the BIL mandate; fidelity is the next problem. Phase I identifies where fidelity is lacking. Phase II supplies it inside the existing pipeline — a corrected rating relationship, not a replacement system.

Scientific

Resolution and complexity thresholds for large-domain inundation modeling remain unsettled in the literature. This produces a quantitative answer, stratified by terrain and forcing regime.

Community

The subgrid property library is contributable through NRDS and deployable through NGIAB. It can be shared within the framework

Extensible

The same architecture accepts spatial data assimilation, extending NextGen's gauge-based EnKF/PF module to satellite flood extent, possible application of Bayesian data assimilation. Subgrid can be used for 2-D surface routing model if BMI controlled model can be adopted into the system.

Thanks!

I am open to research, applied, and regulatory roles.

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Referees and full CV available on request.